Exp no:1

Simulation-Based Electromagnetic Analysis of TEM, TE and TM Propagation Modes in Microwave Transmission Structures

Aim: To simulate and analyze TEM, TE, and TM propagation modes in microwave transmission structures using MATLAB or electromagnetic simulation software, visualize electric-field and magnetic-field distributions, determine cutoff frequency and propagation constant, and compare the propagation characteristics of different modes.

Objective:1. To develop electromagnetic simulation models for TEM, TE and TM propagation modes in
suitable microwave transmission structures.
2. To visualize and analyze the electric-field and magnetic-field distributions associated with
different propagation modes.
3. To determine the cutoff frequency, propagation constant and field characteristics of selected TE
and TM modes.
4. To compare the propagation behavior of different modes and identify their suitability for
practical microwave transmission.

Code

```
clc; clear; close all;

c=3e8; f=10e9; a=22.86e-3; b=10.16e-3;

fc1=c/(2*a); fc2=c/2*sqrt((1/a)^2+(1/b)^2); k=2*pi*f/c;

beta1=sqrt(k^2-(2*pi*fc1/c)^2);

x=linspace(0,a,100); y=linspace(0,b,100); [X,Y]=meshgrid(x,y);

E1=sin(pi*X/a); H1=abs(cos(pi*Y/b)).^4;

E2=sin(pi*X/a).*sin(pi*Y/b);

H2=cos(pi*X/a).*cos(pi*Y/b);

figure('Color','w');

subplot(2,3,1); contourf(X*1e3,Y*1e3,E1,20,'LineColor','none'); colorbar; title('TE10

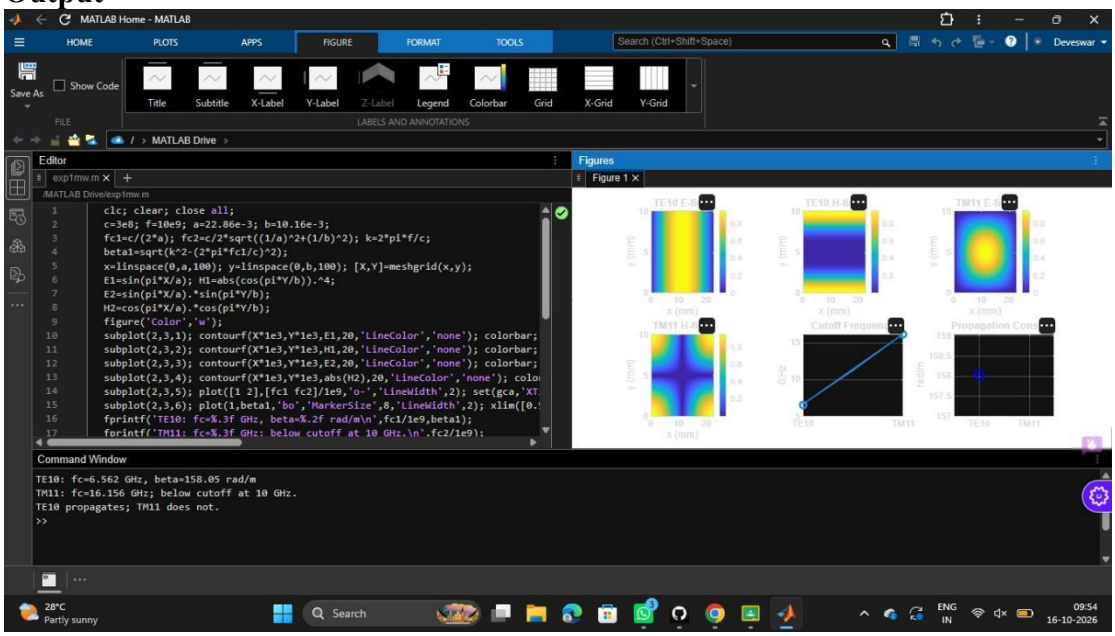
E-field'); xlabel('x (mm)'); ylabel('y (mm)');

subplot(2,3,2); contourf(X*1e3,Y*1e3,H1,20,'LineColor','none'); colorbar; title('TE10

H-field'); xlabel('x (mm)'); ylabel('y (mm)');
```

```
subplot(2,3,3); contourf(X*1e3,Y*1e3,E2,20,'LineColor','none'); colorbar;
title('TM11 E-field'); xlabel('x (mm)'); ylabel('y (mm)');
subplot(2,3,4); contourf(X*1e3,Y*1e3,abs(H2),20,'LineColor','none'); colorbar;
title('TM11 H-field'); xlabel('x (mm)'); ylabel('y (mm)');
subplot(2,3,5); plot([1 2],[fc1 fc2]/1e9,'o-','LineWidth',2); set(gca,'XTick',[1
2],'XTickLabel',{'TE10','TM11'}); ylabel('GHz'); title('Cutoff Frequency'); grid on;
subplot(2,3,6); plot(1,beta1,'bo','MarkerSize',8,'LineWidth',2); xlim([0.5 2.5]);
set(gca,'XTick',[1 2],'XTickLabel',{'TE10','TM11'}); ylabel('rad/m'); title('Propagation
Constant'); grid on;
fprintf('TE10: fc=%.3f GHz, beta=%.2f rad/m\n',fc1/1e9,beta1);
fprintf('TM11: fc=%.3f GHz; below cutoff at 10 GHz.\n',fc2/1e9);
fprintf('TE10 propagates; TM11 does not.\n');
```

Output



$$\beta = \frac{2\pi}{c} f \sqrt{1-\left(\frac{f_c}{f}\right)^2}$$

$$= \frac{2 \times 3.14 \times 10 \times 10^9}{3 \times 10^8} \left[\sqrt{1-\left(\frac{6.65 \times 10^9}{10 \times 10^9}\right)^2}\right.$$

$$= 2.093 \times \sqrt{1-0.4423}$$

$$= 2.093 \times 0.7463$$

$$\beta = 15.620$$

o/p : TE_{10} cutoff = 6.652 GHz

TE_{10} β = 158.05 rad/m

TE_{10} : propagating

TE_{11} : Evanescent (not propagating because $f < f_c$) and h-plane is not possible

$\times 10^3$